10.7 Weyl Ordering and the Midpoint Rule

A classical function $f(q,p)$ does not determine a unique quantum operator: qp could become QP, PQ, or any combination that differs from these by multiples of $[Q,P]=i\hbar$. **Weyl ordering** is the symmetric choice. It is defined on exponentials by

$$e^{xp+kq} \;\longmapsto\; e^{xP+kQ},$$

and extended to any f by writing f as a superposition of exponentials (a Fourier integral). For example $qp \mapsto \tfrac{1}{2}(QP+PQ)$. The function $f(q,p)$ is called the **Weyl symbol** of the operator. Below x and k are parameters; real values give the formulas as written, and $x \to ix$, $k \to ik$ give the unitary version.

Three Orderings of the Same Exponential

Since $[xP, kQ] = xk[P,Q] = -i\hbar xk$ is a c-number, the Baker–Campbell–Hausdorff formula $e^A e^B = e^{A+B+\frac{1}{2}[A,B]}$ is *exact*:

$$e^{xP}e^{kQ} = e^{xP+kQ}\,e^{-i\hbar xk/2}, \qquad e^{kQ}e^{xP} = e^{xP+kQ}\,e^{+i\hbar xk/2}.$$

The phases $e^{\mp i\hbar xk/2}$ are the ordering dependence. The symmetric splitting has none: apply BCH twice,

$$e^{\frac{kQ}{2}}e^{xP}e^{\frac{kQ}{2}} = e^{\frac{kQ}{2}}\,e^{xP+\frac{kQ}{2}}\,e^{-i\hbar xk/4} = e^{xP+kQ}\,e^{+i\hbar xk/4}\,e^{-i\hbar xk/4} = e^{xP+kQ}.$$

This is an identity, not an approximation: the Weyl-ordered exponential *is* the symmetric product. The same holds with P split instead, $e^{xP/2}e^{kQ}e^{xP/2} = e^{xP+kQ}$.

The Midpoint Rule

Take matrix elements between position eigenstates, with $\langle q|p\rangle = e^{ipq/\hbar}/\sqrt{2\pi\hbar}$. The outer factors act on the bra and ket and give $e^{kq'/2}$ and $e^{kq/2}$:

$$\langle q'|e^{xP+kQ}|q\rangle = e^{k(q+q')/2}\,\langle q'|e^{xP}|q\rangle = \int \frac{dp}{2\pi\hbar}\,e^{ip(q'-q)/\hbar}\,e^{xp+k\bar{q}}, \qquad \bar{q}=\frac{q+q'}{2}.$$

The classical function e^{xp+kq} appears evaluated at the *midpoint* $\bar{q}$. Because every Weyl-ordered operator is a superposition of such exponentials, the result holds for all of them:

$$\langle q'|f_W(Q,P)|q\rangle = \int \frac{dp}{2\pi\hbar}\,e^{ip(q'-q)/\hbar}\,f(\bar{q},p).$$

Check with $f=qp$: $\langle q'|\frac{1}{2}(QP+PQ)|q\rangle = \frac{1}{2}(q'+q)\langle q'|P|q\rangle$, which is the midpoint value times the matrix element of P.

Connection to the Path Integral

Slice the time evolution into steps Δt and insert position and momentum eigenstates. If the Hamiltonian operator is written in Weyl-ordered form with symbol $H_W(q,p)$, each short-time matrix element is given by the midpoint rule, and the phase-space path integral reads

$$\int \mathcal{D}q\,\mathcal{D}p\ \exp\Big\{\frac{i}{\hbar}\int dt\,\big[p\dot{q}-H_W(q,p)\big]\Big\},$$

with H_W evaluated at $\bar{q}_j=\frac{1}{2}(q_j+q_{j+1})$ on each slice.

- **The midpoint rule selects an ordering; it does not remove the ambiguity.** A discretization of the path integral corresponds to an ordering of the operator H. Different choices give Hamiltonians that differ by terms of order $\hbar$, and the midpoint rule is the one that corresponds to Weyl ordering.
- **For $H=\frac{P^2}{2m}+V(Q)$ there is nothing to choose.** No term mixes Q and P, H_W is the classical Hamiltonian, and evaluating V at q_j, q_{j+1} or $\bar{q}_j$ gives the same continuum limit.
- **When Q and P mix, the choice matters.** For a charged particle, $H=(P-eA(Q))^2/2m$ contains $PA+AP$, which is Weyl ordered; the $\dot{q}\cdot A$ term must be evaluated at the midpoint to obtain the gauge-invariant Schrödinger equation.

Weyl Ordering Is Not the Trotter Splitting

A different splitting is used to build the path integral in the first place. For $H=K+V$ with $K=P^2/2m$,

$$e^{-iH\Delta t/\hbar}=e^{-iK\Delta t/2\hbar}\,e^{-iV\Delta t/\hbar}\,e^{-iK\Delta t/2\hbar}+\mathcal{O}(\Delta t^3),$$

the symmetric Trotter (Strang) splitting. It is an approximation, exact only as $\Delta t\to 0$, because $[K,V]$ is an operator and higher BCH terms do not vanish; the errors, $\mathcal{O}(\Delta t^3)$ per step, sum to zero over the $T/\Delta t$ steps. The identity $e^{kQ/2}e^{xP}e^{kQ/2}=e^{xP+kQ}$ is different: it is exact, because $[Q,P]$ is a c-number.

Summary

- Weyl ordering maps e^{xp+kq} to e^{xP+kQ}, and qp to $\frac{1}{2}(QP+PQ)$.
- $e^{kQ/2}e^{xP}e^{kQ/2}=e^{xP+kQ}$ exactly; the non-symmetric orderings differ from it by the phases $e^{\mp i\hbar xk/2}$.
- Matrix elements of Weyl-ordered operators put the classical function at the midpoint $\bar{q}$. In the path integral, the midpoint rule is the discretization that corresponds to Weyl ordering; for $P^2/2m+V(Q)$ any evaluation point gives the same continuum limit.